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$d = r_{Na^+} + r_{F^-} \quad ---(5)$

i.e. $r_{Na^+} + r_{F^-} = 231\ pm$

We know that

$$\frac{r_{Na^+}}{r_{F^-}} = \frac{(Z_{eff})_{F^-}}{(Z_{eff})_{Na^+}}$$

$(Z_{eff})_{F^-} = Z - S$

$= 9 - 4.15$

$= 4.85$

$(Z_{eff})_{Na^+} = 11 - 4.15$

$= 6.85$

$$\therefore \frac{r_{Na^+}}{r_{F^-}} = \frac{4.85}{6.85}$$

$= 0.71$

$\Rightarrow r_{Na^+} = 0.71 \times r_{F^-}$

Substituting (3) in (1)

(1) $\Rightarrow$ $0.71\, r_{F^-} + r_{F^-} = 231\ pm$

$1.71\, r_{F^-} = 231\ pm$

$$r_{F^-} = \frac{231}{1.71} = 135.1\ pm$$

Substituting the value of r_{F^-} in equation (1)

$r_{Na^+} + 135.1 = 231$

$r_{Na^+} = 95.9\ pm$

Evaluate Yourself

5. A student reported the ionic radii of isoelectronic species X^{3+}, Y^{2+} and Z^- as 136 pm, 64 pm and 49 pm respectively. Is that order correct? Comment.

3.5.3 Ionisation energy

It is defined as the minimum amount of energy required to remove the most loosely bound electron from the valence shell of the isolated neutral gaseous atom in its ground state. It is expressed in kJ mol^{-1} or in electron volts (eV).

$$M_{(g)} + IE_1 \rightarrow M^+_{(g)} + 1\ e^-$$

Where IE_1 represents the first ionisation energy.

Successive Ionisation energies

The minimum amount of energy required to remove an electron from a unipositive cation is called second ionisation energy. It is represented by the following equation.

$$M^+_{(g)} + IE_2 \rightarrow M^{2+}_{(g)} + 1\ e^-$$

In this way we can define the successive ionisation energies such as third, fourth etc.

The total number of electrons are less in the cation than the neutral atom while the nuclear charge remains the same. Therefore the effective nuclear charge of the cation is higher than the corresponding neutral atom. Thus the successive ionisation energies, always increase in the following order

$$IE_1 < IE_2 < IE_3 < .....$$

Periodic Trends in Ionisation Energy

The ionisation energy usually increases along a period with few exceptions. As discussed earlier, when we move from left to right along a period, the valence electrons are added to the same shell, at the same time protons are added to the nucleus. This successive increase of nuclear charge increases the electrostatic attractive force on the valence electron and more energy is required to remove the valence electron resulting in high ionisation energy.

Let us consider the variation in ionisation energy of second period